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$$\int \frac{x^3}{x^2 - 1} dx$$
$$= \int \left(x + \frac{x}{x^2 - 1} \right) dx$$

stel $u = x^2 - 1$

$$du = 2x dx$$

$$= \frac{x^2}{2} + \frac{1}{2} \int \frac{du}{u}$$
$$= \frac{x^2}{2} + \frac{1}{2} \ln |u| + C$$
$$= \frac{x^2}{2} + \frac{1}{2} \ln |x^2 - 1| + C$$

References